

Your Paper's Title(SHUMOJIAYOUZHAN)

Summary

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Contents

1	Introduction	3
1.1	Problem Background	3
1.2	Problem Restatement	3
1.3	Our Work	5
2	Preparation of the Models	5
2.1	Assumptions	5
2.2	Notation	5
2.3	Data Preparation	5
3	Model	7
3.1	Model 1	7
3.1.1	Mass Requirement	9
3.1.2	Space Elevator Capacity Constraint	9
3.1.3	Rocket Launch Frequency Constraint (Earth-based)	9
3.1.4	Orbital Rocket Fleet Throughput (Apex-to-Moon Cycling)	10
3.1.5	Fleet Sizing and Safety Fleet Threshold I_{safe}	10
3.1.6	Scenario-Specific Delivered Mass	11
3.1.7	Cost Model (for Optimization or Result Reporting)	11
3.2	Closed-Form Solutions for Scenario A and Scenario B	12
3.2.1	Scenario A: Pure Space Elevator Scenario	12
3.2.2	Scenario B: Pure Rocket Scenario	13
3.3	Optimization of the Hybrid Transportation Scenario	14
3.3.1	Decision Variable	14
3.3.2	Delivered Mass Decomposition	14
3.3.3	Effective Transportation Rate and Project Duration	14
3.3.4	Cost Model	15
3.3.5	Weighted-Sum Multi-Objective Optimization	15

3.3.6	Feasible Range of the Decision Variable	16
3.4	Model 2: Reliability Model under Non-Ideal Operations	16
3.4.1	Motivation and Modeling Scope	16
3.4.2	Baseline Parameter Inheritance from Task 1	16
3.4.3	Program Decomposition and Failure-State Definition	16
3.4.4	Scenario-Dependent Nominal Program Rule	17
3.4.5	Absorbing Markov Transition Model for Rocket Failures	17
3.4.6	Failure Probability Synthesis and Parameter Mapping	18
3.4.7	Apex Inventory Coupling for Elevator-Supported Scenarios	19
3.4.8	Event-Driven Discrete-Event Simulation (DES)	20
3.4.9	Closed-Loop Replacement with Latency and Launch Cadence	21
3.4.10	DES Routine with Inventory and Replacement	21
3.4.11	Monte Carlo Protocol, Outputs, and Verification Checks	22
4	Validating the Model	23
5	Conclusions	23
6	A Summary	23
7	Evaluate of the Mode	23
8	Strengths and weaknesses	23
8.1	Strengths	24
	Appendices	24
	Appendix A First appendix	24
	Appendix B Second appendix	25

1 Introduction

1.1 Problem Background

As human civilization expands, Earth's fragile ecosystems face severe pressures from population surges, resource depletion, and environmental degradation, driving a resurgence in space exploration. The Moon, with its proximity advantage and potential as a gateway to deep space, emerges as the foremost candidate for permanent settlement. The space elevator system, a revolutionary infrastructure, utilizes electrically powered graphene cables (with a strength of 130 GPa) to efficiently transport materials from equatorial ground ports to geostationary orbit (GEO) and the top anchor point. Each port has an annual transport capacity of 179,000 metric tons, with three ports synergistically reaching 537,000 metric tons—reducing fuel consumption and pollution by 90%. The Moon Colonization Management (MCM) agency plans to launch a 100,000-person base by 2050, requiring 100 million tons of materials and sustained water supply. Traditional rockets (with an annual transport capacity of only 78,000 tons) would take millennia to accomplish this, while the space elevator could reduce the timeline to decades. However, challenges in cable maintenance and system reliability must be addressed. This paper evaluates pure elevator, pure rocket, and hybrid solutions using Markov failure analysis to assess cost, time, reliability, and environmental impact, providing MCM with optimal logistics decision-making.

1.2 Problem Restatement

The proposed construction of a large-scale lunar colony represents a critical step in humanity's long-term expansion into space, yet it also introduces substantial logistical, economic, and environmental challenges. The status quo of Earth-to-Moon transportation technologies demands a systematic solution to balance construction efficiency, long-term sustainability, and environmental responsibility. Based on a comprehensive analysis of the problem context, the objectives of this study are restated as follows:

- Construct a mathematical model to evaluate the cost and time efficiency of transporting materials from Earth to a lunar colony using different transportation schemes, including space elevator systems, traditional rocket launches, and hybrid approaches. The model should account for large-scale construction demands as well as ongoing supply requirements.
- Incorporate system uncertainties and operational disruptions into the model, such as partial failures of space elevator components or rocket launch inefficiencies, and analyze how these factors affect overall transportation performance, cost, and project timelines.
- Perform a sensitivity analysis to identify the most influential parameters in the transportation system, including payload capacity, transportation rate, failure probability, and environmental impact, and evaluate the robustness of the proposed solutions under varying assumptions.

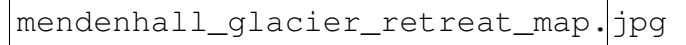
A rectangular placeholder box for a map showing the retreat of Mendenhall Glacier.A rectangular placeholder box for a chart showing tourism-related concerns from Juneau residents.

Figure 1: Retreat of Mendenhall Glacier and tourism-related concerns from Juneau residents

- Extend the model to post-construction operations by estimating the annual water demand for a fully operational lunar colony of 100,000 residents and analyzing the additional cost and time required to sustain the population during its first year of habitation.
- Communicate the results through a policy-oriented recommendation, summarizing model outcomes, comparing alternative transportation strategies, and proposing an optimal plan to support the construction and long-term operation of the lunar colony while minimizing adverse environmental effects on Earth.

1.3 Our Work

2 Preparation of the Models

2.1 Assumptions

2.2 Notation

2.3 Data Preparation

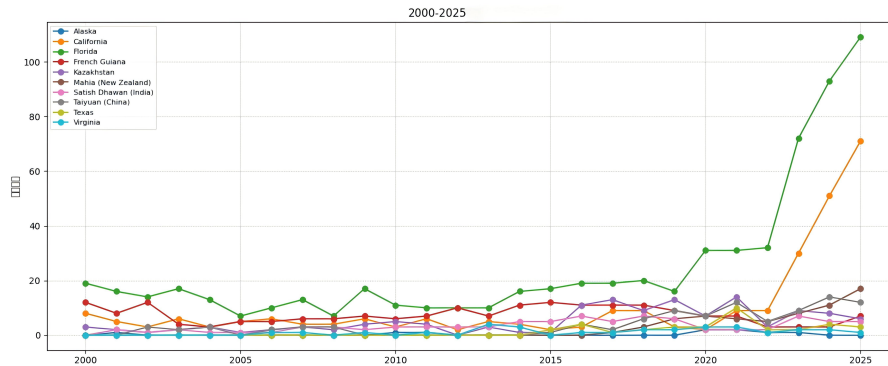


Figure 2: aa

To model the long-term reduction in launch cost, an experience-curve-based exponential decay model is adopted. In the aerospace industry, a well-established empirical rule suggests that unit cost decreases by approximately 15–20% with each doubling of cumulative production. This learning effect is captured by the following formulation:

$$Cost(t) = C_0 \exp[-k(t - t_0)], \quad (1)$$

where C_0 denotes the baseline expendable launch cost and k represents the learning rate associated with increased production scale and operational maturity. Using a reference cost of $C_0 = \$150$ million and calibrating the model over a 24-year horizon, the parameter k is selected as $k \approx 0.096$, which lies within the commonly accepted range $0.05 \leq k \leq 0.10$. Under this calibration, the model yields an average single-launch cost of approximately \$15 million, corresponding to an order-of-magnitude reduction relative

Table 1: Notations and Definitions

Symbol	Definition	Unit	Value/Est.
Decision Variables			
M_{SE}	Total mass transported via Space Elevator system	ton	≥ 0
N_{Rock}	Total number of Falcon Heavy launches	-	Integer
T	Total project duration (Time-to-Completion)	year	≥ 0
Global Parameters			
M_{goal}	Total material mass required for the Moon Colony	ton	10^8
Cost & Capacity Parameters			
C_{launch}	Cost per single Falcon Heavy launch	USD	1.5×10^8
C_{elec_unit}	Electricity cost per ton via Space Elevator	USD/ton	4.15^*
C_{maint}	Annual maintenance cost for SE system	USD/yr	1.2×10^8
C_{TV_fixed}	Fixed cost for Apex-to-Moon Transfer system	USD/yr	3.0×10^8
Rate Constraints			
Cap_{SE}	Annual capacity of the Space Elevator system	ton/yr	5.37×10^5
Cap_{Rock}	Payload capacity of a single rocket launch	ton	125
f_{total}	Max. annual Earth launch frequency (all sites combined)	launches/yr	Data-driven
f_{cycle}	Max. round-trips per rocket per year (apex-to-Moon)	cycles/yr	$\approx 60^\dagger$

*Note: C_{elec_unit} is derived from the weekly energy consumption (E_{week}) and annual elevator throughput.

† Note: f_{cycle} corresponds to a 6-day round trip cycle (approximately 365/6 per year).

Table 2: Additional Notations for Task 2 (Non-Ideal Conditions)

Symbol	Definition	Unit	Value/Est.
Stochastic Programs & Markov Model			
$\mathcal{L}$	Set of operational programs $\{1, 2, 3, 4, 5\}$	-	Defined
6	Absorbing failure state (rocket removed permanently)	-	Defined
$\bar{\mathcal{L}}$	Extended state space $\mathcal{L} \cup \{6\}$	-	Defined
$\phi_h(l)$	Nominal next-program rule under scenario h	-	Task 1
p_{l6}	Failure probability during program l (transition to state 6)	-	TBD
$\bar{\mathbf{P}}^{(h)}$	Markov transition matrix under scenario h on $\bar{\mathcal{L}}$	-	Derived
$L(t)$	Program/state of a rocket at time t	-	Simulated
Robust Scheduling & Replacement			
$I(t)$	Number of active rockets at time t	rockets	Simulated
I_{safe}	Safety threshold for active fleet size	rockets	From Task 1
$R(t)$	Replacement (relaunch) decision at time t	rockets	Policy
δ	Replacement latency (delay before replacement becomes available)	day/yr	TBD
$\Delta\tau$	Time buffer added to absorb disruptions	day/yr	Tuned
Failure Loss (Optional)			
c_l^{loss}	Cost/loss incurred if failure occurs during program l	USD	TBD

Table 3: Launch-Site-Level Forecast and Average Launch Interval in 2050 (Aggressive Extrapolation Scenario)

Launch Site	2025	2050 Forecast	Avg. Interval	Unit
Alaska (USA)	0	10	~36.5	days
Florida (USA)	109	2300	~0.16	days
California (USA)	71	950	~0.38	days
Texas (USA)	3	190	~1.92	days
Virginia (USA)	1	70	~5.2	days
Mahia (New Zealand)	17	140	~2.6	days
Taiyuan (China)	12	100	~3.7	days
Satish Dhawan (India)	5	35	~10.4	days
Kazakhstan (Baikonur)	6	25	~14.6	days
French Guiana (Kourou)	7	24	~15.2	days

Note: Average launch interval is calculated as 365 divided by the annual launch frequency. Values below one day indicate multiple launches per day on average. This table reflects an upper-bound scenario assuming unconstrained high growth.

to the expendable baseline. This value is consistent with an aggressive scenario characterized by full reusability, high launch cadence, and mature operational efficiency, and is therefore adopted as the model-implied marginal launch cost in subsequent analyses.

Table 4: Falcon Heavy–Class Launch Cost Model Inputs (Assumption Set)

Parameter	Real-World Reference	Model Input	
Single-launch cost	~ \$150M (expendable baseline)	\$15M (average)	Assumes fully mature
Payload per launch	~20 t (TLI reference)	125 t (average)	Strictly follows
Reuse lifetime	< 20 flights	100 flights	Enabled by materials/
Propellant type	RP-1 (kerosene)	Methalox (CH₄/LOX)	Lower coking ⇒ l

3 Model

3.1 Model 1

Scenario A: Space Elevator–Dominated Transportation System (Pure Elevator Scenario)

In this scenario, all construction materials required for the lunar settlement are first transported from the Earth’s surface to the orbital anchor point via the Galactic Harbour space elevator system. The elevator provides a continuous and pollution-free logistics channel from Earth to orbit, operating at a fixed annual throughput determined by its design capacity.

After the initial deployment phase, transport rockets are permanently stationed in orbit and no longer return to Earth. Each rocket is initially launched from the Earth’s surface without payload and transferred to the orbital anchor point, where it enters a repetitive

operational cycle. In each cycle, a rocket loads materials delivered by the space elevator at the anchor point, transports them to the lunar settlement, unloads the payload, and then returns to the anchor point for the next loading operation. All subsequent material transfers between Earth and the Moon are therefore completed through the combined operation of the space elevator (Earth to anchor point) and the orbital rocket fleet (anchor point to Moon).

Under this scenario, rockets are used exclusively for cislunar transportation and do not perform repeated Earth surface launches. As a result, the transportation system avoids the high fuel consumption and environmental impact associated with frequent atmospheric launches. The overall delivery rate to the Moon is jointly constrained by the elevator's upward transport capacity and the total carrying capacity of the orbital rocket fleet.

Scenario B: Conventional Rocket-Based Transportation System (Pure Rocket Scenario)

In the pure rocket scenario, all materials required for lunar base construction are transported directly from the Earth's surface to the Moon using conventional launch vehicles. Each transportation cycle consists of a fully loaded rocket launch from a ground-based launch site, insertion into a translunar trajectory, delivery of payload to the lunar settlement, and a return flight to the original launch site.

Rockets repeatedly execute the complete launch–transport–return cycle throughout the project duration until all required materials have been delivered. No orbital infrastructure, such as a space elevator or an anchor-point logistics hub, is utilized in this scenario. Consequently, every transportation task requires overcoming Earth's gravitational well from the surface, leading to substantial fuel consumption and high operational costs per delivery.

While this approach benefits from a relatively simple system architecture and does not require the construction or maintenance of orbital facilities, its overall transportation efficiency is limited by launch frequency, fuel requirements, and the cumulative cost of repeated Earth-based launches. The total delivery rate to the lunar settlement is determined by the annual number of launches and the payload capacity of each rocket.

Scenario C: Hybrid Space Elevator–Rocket Transportation System (Mixed Scenario)

The hybrid scenario integrates the advantages of both the space elevator and conventional rockets by dividing the transportation process into two operational phases. In the initial phase, rockets are launched from the Earth's surface with full payloads and deliver materials directly to the lunar settlement. After completing this first delivery, instead of returning to Earth, these rockets are transferred to the orbital anchor point and retained in orbit as part of a permanent transportation fleet.

In the second phase, the retained rockets operate in a closed-loop cislunar logistics cycle. Materials are continuously transported from the Earth's surface to the orbital anchor point by the space elevator, where they are loaded onto the orbiting rockets. Each

rocket then transports the payload from the anchor point to the Moon and returns to the anchor point empty, ready for the next cycle. This process is repeated until all remaining materials required for lunar base construction have been delivered.

By allowing rockets to perform a single Earth-based launch before transitioning to orbital operations, the hybrid scenario reduces the number of costly and fuel-intensive surface launches while maintaining a flexible and scalable transportation capacity. The system's performance depends on the balance between the space elevator's supply rate and the number of rockets retained in orbit, enabling a trade-off between total transportation time and overall launch cost.

The following are definitions for various status:

- (1) Launch the empty rocket from the ground to the top anchor point of Galaxy Port, or transport rocket components via the Galaxy Port elevator to the top anchor point for assembly;
- (2) Launch the fully loaded rocket from the ground into the Earth-Moon transfer orbit;
- (3) Depart the fully loaded rocket from the Earth-Moon transfer orbit to the lunar colony for unloading;
- (4) Empty rocket is retrieved from the lunar colony and returned to the top anchor point of Galaxy Port;
- (5) Empty rocket is retrieved from the Earth-Moon transfer orbit and returned to its original launch site.

Based on the above fundamental procedures, the three transport scenarios can be formally described from the operational perspective of a single rocket:

3.1.1 Mass Requirement

The total mass required for constructing the lunar colony is

$$M_{goal} = 10^8 \text{ ton.} \quad (2)$$

Any feasible transportation plan must satisfy the total delivered mass constraint:

$$M_{delivered} \geq M_{goal}. \quad (3)$$

3.1.2 Space Elevator Capacity Constraint

The total mass transported by the Space Elevator system is bounded by its annual capacity:

$$M_{SE} \leq Cap_{SE} \cdot T. \quad (4)$$

3.1.3 Rocket Launch Frequency Constraint (Earth-based)

Given the maximum annual launch frequency across all Earth launch sites,

$$N_{Rock} \leq f_{total} \cdot T. \quad (5)$$

3.1.4 Orbital Rocket Fleet Throughput (Apex-to-Moon Cycling)

Assuming each launched rocket can be retained to operate in orbit and perform at most f_{cycle} round trips per year (Apex $\leftrightarrow$ Moon), the upper bound of Apex-to-Moon delivered mass is

$$M_{\text{Apex} \rightarrow \text{Moon}} \leq N_{\text{Rock}} \cdot f_{cycle} \cdot Cap_{\text{Rock}} \cdot T. \quad (6)$$

3.1.5 Fleet Sizing and Safety Fleet Threshold I_{safe}

In Task 1 (ideal operation), the transportation system is sized to sustain a target steady delivery rate without long-term queueing or capacity collapse. This yields a *safety fleet threshold* I_{safe} , which will be inherited by Task 2 as the baseline fleet size for closed-loop replenishment.

General sizing principle. Let $q = Cap_{\text{Rock}}$ denote payload per successful delivery (ton), and let $T_{\text{cycle}}^{(h)}$ be the ideal cycle time (year) for one rocket under scenario h , defined as the sum of ideal program durations along one complete loop. If the system is required to sustain a long-run delivery rate $R_{\text{target}}^{(h)}$ (ton/year), then the minimum fleet size that can support this rate is

$$I_{\text{safe}}^{(h)} = \left\lceil \frac{R_{\text{target}}^{(h)} T_{\text{cycle}}^{(h)}}{q} \right\rceil. \quad (7)$$

Scenario-specific $R_{\text{target}}^{(h)}$ and $T_{\text{cycle}}^{(h)}$. For the elevator-supported cycling (Scenario A and the steady phase of Scenario C), the natural target is to *match the elevator supply*:

$$R_{\text{target}}^{(A)} = Cap_{SE}. \quad (8)$$

In addition, one cislunar loop corresponds to programs $(3) \rightarrow (4) \rightarrow (3)$, thus

$$T_{\text{cycle}}^{(A)} \approx \tau_3 + \tau_4, \quad \Rightarrow \quad I_{\text{safe}}^{(A)} = \left\lceil \frac{Cap_{SE}(\tau_3 + \tau_4)}{Cap_{\text{Rock}}} \right\rceil. \quad (9)$$

Equivalently, using $f_{cycle} \approx 1/(\tau_3 + \tau_4)$ (cycles/year), we recover the Task 1 steady-state matching condition:

$$I_{\text{safe}}^{(A)} = \left\lceil \frac{Cap_{SE}}{f_{cycle} Cap_{\text{Rock}}} \right\rceil. \quad (10)$$

For the Earth-launched cycling (Scenario B), the system throughput is limited by the maximum annual launch cadence. A natural target is the *maximum achievable* delivery rate:

$$R_{\text{target}}^{(B)} = f_{\text{total}} Cap_{\text{Rock}}. \quad (11)$$

One full Earth-based loop is $(2) \rightarrow (3) \rightarrow (5) \rightarrow (2)$, hence

$$T_{\text{cycle}}^{(B)} \approx \tau_2 + \tau_3 + \tau_5, \quad \Rightarrow \quad I_{\text{safe}}^{(B)} = \lceil f_{\text{total}} (\tau_2 + \tau_3 + \tau_5) \rceil. \quad (12)$$

For the hybrid scenario (Scenario C), we use the same sizing principle in the steady phase:

$$I_{\text{safe}}^{(C)} \approx I_{\text{safe}}^{(A)} = \left\lceil \frac{Cap_{SE}}{f_{\text{cycle}} Cap_{Rock}} \right\rceil, \quad (13)$$

while the initial deployment phase is handled by the launch-cadence constraint as in Task 1. The resulting I_{safe} values are treated as *baseline fleet thresholds* in Task 2.

3.1.6 Scenario-Specific Delivered Mass

We formalize the three scenarios via different expressions of $M_{\text{delivered}}$:

Scenario A: Pure Elevator Scenario (Elevator-dominated) All cargo first reaches the orbital anchor point via the Space Elevator, thus the Moon delivery is limited by both the elevator supply and orbital cycling capability:

$$M_{\text{delivered}}^{(A)} \leq \min \left(M_{SE}, N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock} \cdot T \right). \quad (14)$$

The feasibility requirement becomes

$$\min \left(M_{SE}, N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock} \cdot T \right) \geq M_{\text{goal}}. \quad (15)$$

Scenario B: Pure Rocket Scenario (Earth-to-Moon direct launches) All cargo is delivered by Earth-based launches. Each launch delivers at most Cap_{Rock} , therefore

$$M_{\text{delivered}}^{(B)} \leq N_{Rock} \cdot Cap_{Rock}. \quad (16)$$

The feasibility requirement becomes

$$N_{Rock} \cdot Cap_{Rock} \geq M_{\text{goal}}. \quad (17)$$

Scenario C: Hybrid Scenario (One-time direct + orbital cycling with elevator supply) In the hybrid design, each rocket performs one Earth-to-Moon delivery at deployment, and then participates in the Apex-to-Moon cycling supported by elevator supply. Hence, the total delivered mass can be bounded by

$$M_{\text{delivered}}^{(C)} \leq N_{Rock} \cdot Cap_{Rock} + \min \left(M_{SE}, N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock} \cdot T \right). \quad (18)$$

The feasibility requirement becomes

$$N_{Rock} \cdot Cap_{Rock} + \min \left(M_{SE}, N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock} \cdot T \right) \geq M_{\text{goal}}. \quad (19)$$

3.1.7 Cost Model (for Optimization or Result Reporting)

The total cost consists of launch cost, elevator electricity cost, annual maintenance, and a fixed transfer-system cost:

$$C_{\text{total}} = C_{\text{launch}} \cdot N_{Rock} + C_{\text{elec_unit}} \cdot M_{SE} + C_{\text{maint}} \cdot T + C_{TV_fixed}. \quad (20)$$

3.2 Closed-Form Solutions for Scenario A and Scenario B

Scenarios A and B admit direct analytical solutions because their logistics chains contain no internal trade-off once feasibility is enforced. In this section, we derive the minimum completion time and the corresponding resource requirements under each scenario.

3.2.1 Scenario A: Pure Space Elevator Scenario

In Scenario A, all payload must be transported from the Earth surface to the orbital anchor point by the Space Elevator system, and then delivered to the Moon by the orbital rocket fleet. Therefore, the elevator-delivered mass must cover the entire requirement:

$$M_{SE} = M_{goal}. \quad (21)$$

The Space Elevator capacity constraint implies

$$M_{SE} \leq Cap_{SE} \cdot T, \quad (22)$$

which yields the elevator-limited lower bound on completion time:

$$T_{\text{elev}}^{(A)} = \frac{M_{goal}}{Cap_{SE}}. \quad (23)$$

However, Scenario A also requires an initial deployment phase in which rockets are launched *empty* from Earth to the apex anchor. Given the maximum annual launch frequency f_{total} , deploying N_{Rock} rockets requires at least

$$T_{\text{deploy}}^{(A)} = \frac{N_{Rock}}{f_{total}}. \quad (24)$$

Therefore, the total project duration in Scenario A is approximated by the slower process between elevator supply and rocket deployment:

$$T_A^* = \max\left(T_{\text{elev}}^{(A)}, T_{\text{deploy}}^{(A)}\right) = \max\left(\frac{M_{goal}}{Cap_{SE}}, \frac{N_{Rock}}{f_{total}}\right). \quad (25)$$

To prevent long-term accumulation of cargo at the anchor point, the orbital rocket fleet must be able to transport at least as much as the elevator supplies over the project horizon. Using the rocket cycling limit f_{cycle} and per-trip payload Cap_{Rock} , the cumulative Apex-to-Moon transport capacity is

$$M_{\text{Apex} \rightarrow \text{Moon}} \leq N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T. \quad (26)$$

Feasibility requires $M_{\text{Apex} \rightarrow \text{Moon}} \geq M_{SE}$, hence

$$N_{Rock} \cdot f_{cycle} \cdot Cap_{Rock} \cdot T_A^* \geq M_{goal}. \quad (27)$$

Substituting $T_A^* = M_{goal}/Cap_{SE}$, the required number of rockets is bounded by

$$N_{Rock} \geq \frac{Cap_{SE}}{f_{cycle} \cdot Cap_{Rock}}, \quad (28)$$

and thus the minimal integer solution is

$$N_{Rock,A}^* = \left\lceil \frac{Cap_{SE}}{f_{cycle} \cdot Cap_{Rock}} \right\rceil. \quad (29)$$

Finally, the total cost under Scenario A can be evaluated by the unified cost model:

$$C_A = C_{launch} \cdot N_{Rock,A}^* + C_{elec_unit} \cdot M_{SE} + C_{maint} \cdot T_A^* + C_{TV_fixed}, \quad (30)$$

where $M_{SE} = M_{goal}$ and $T_A^* = M_{goal}/Cap_{SE}$.

3.2.2 Scenario B: Pure Rocket Scenario

In Scenario B, all payload is delivered directly from the Earth surface to the Moon by conventional rocket launches. Therefore, the total delivered mass is bounded by

$$M_{delivered}^{(B)} \leq N_{Rock} \cdot Cap_{Rock}. \quad (31)$$

Feasibility requires

$$N_{Rock} \cdot Cap_{Rock} \geq M_{goal}, \quad (32)$$

which gives the minimum number of launches

$$N_{Rock,B}^* = \left\lceil \frac{M_{goal}}{Cap_{Rock}} \right\rceil. \quad (33)$$

Given the maximum annual launch frequency f_{total} (all sites combined), the launch-count constraint is

$$N_{Rock} \leq f_{total} \cdot T. \quad (34)$$

Hence, the minimum feasible completion time is obtained by saturating the launch rate:

$$T_B^* = \frac{N_{Rock,B}^*}{f_{total}}. \quad (35)$$

The corresponding total cost under Scenario B is

$$C_B = C_{launch} \cdot N_{Rock,B}^*. \quad (36)$$

Note that Scenario B does not require Space Elevator electricity costs or the fixed Apex-to-Moon transfer infrastructure cost because no orbital logistics hub is utilized.

3.3 Optimization of the Hybrid Transportation Scenario

Unlike Scenarios A and B, which admit direct analytical solutions, Scenario C (the Hybrid Space Elevator–Rocket Transportation System) involves an inherent trade-off between transportation time and total system cost. Increasing the number of deployed rockets accelerates material delivery but incurs higher launch and operational costs. Therefore, Scenario C is formulated as a bi-objective optimization problem.

3.3.1 Decision Variable

The key decision variable in Scenario C is the total number of rockets launched from Earth and subsequently retained for orbital cycling:

$$N_{Rock} \in \mathbb{Z}_{\geq 0}. \quad (37)$$

Each rocket performs one initial fully loaded Earth-to-Moon delivery during the deployment phase and then joins the Apex-to-Moon cyclic transportation supported by the Space Elevator system.

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3.3.2 Delivered Mass Decomposition

The total mass delivered directly to the Moon during the deployment phase is

$$M_{\text{direct}} = N_{Rock} \cdot Cap_{Rock}. \quad (38)$$

The remaining mass must be supplied via the Space Elevator system:

$$M_{SE} = \max(0, M_{\text{goal}} - N_{Rock} \cdot Cap_{Rock}). \quad (39)$$

—

3.3.3 Effective Transportation Rate and Project Duration

After deployment, the orbital rocket fleet provides an annual Apex-to-Moon transportation capacity of

$$R_{\text{rocket}} = N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock}. \quad (40)$$

The Space Elevator system provides an annual supply capacity of Cap_{SE} . Hence, the effective annual delivery rate during the steady-state phase is

$$R(N_{Rock}) = \min(Cap_{SE}, N_{Rock} \cdot f_{\text{cycle}} \cdot Cap_{Rock}). \quad (41)$$

The time required to complete delivery of the remaining mass is approximated by

$$T_{\text{remain}}(N_{Rock}) = \frac{M_{SE}}{R(N_{Rock})}. \quad (42)$$

In addition, deploying N_{Rock} rockets from Earth is limited by the maximum annual launch frequency f_{total} , resulting in a deployment time of

$$T_{\text{deploy}}(N_{Rock}) = \frac{N_{Rock}}{f_{total}}. \quad (43)$$

The total project duration is approximated as

$$T(N_{Rock}) = \max(T_{\text{deploy}}(N_{Rock}), T_{\text{remain}}(N_{Rock})). \quad (44)$$

—

3.3.4 Cost Model

The total system cost for Scenario C consists of rocket launch costs, Space Elevator electricity costs, annual maintenance costs, and a fixed transfer system cost:

$$C(N_{Rock}) = C_{\text{launch}} \cdot N_{Rock} + C_{\text{elec_unit}} \cdot M_{SE} + C_{\text{maint}} \cdot T(N_{Rock}) + C_{TV_fixed}. \quad (45)$$

—

3.3.5 Weighted-Sum Multi-Objective Optimization

Scenario C is formulated as a bi-objective minimization problem with objectives of minimizing total project duration and total cost:

$$\min_{N_{Rock}} (T(N_{Rock}), C(N_{Rock})). \quad (46)$$

To obtain Pareto-optimal solutions, the weighted-sum method is adopted. The normalized scalar objective function is defined as

$$J(N_{Rock}) = \alpha \frac{T(N_{Rock})}{T_{\text{ref}}} + (1 - \alpha) \frac{C(N_{Rock})}{C_{\text{ref}}}, \quad \alpha \in [0, 1], \quad (47)$$

where T_{ref} and C_{ref} are reference values used for normalization, chosen as the maximum observed project duration and cost among all feasible solutions.

By varying the weight parameter α , a set of Pareto-optimal solutions is generated, representing different trade-offs between time efficiency and economic cost. The final recommended solution may be selected based on practical preferences, such as a balanced compromise near the knee point of the Pareto frontier.

—

3.3.6 Feasible Range of the Decision Variable

To ensure meaningful system operation, the search range of N_{Rock} is bounded as follows:

$$\left\lceil \frac{Cap_{SE}}{f_{cycle} \cdot Cap_{Rock}} \right\rceil \leq N_{Rock} \leq \left\lceil \frac{M_{goal}}{Cap_{Rock}} \right\rceil. \quad (48)$$

The lower bound guarantees that the orbital rocket fleet is capable of matching the Space Elevator supply rate, while the upper bound corresponds to delivering all required mass through direct Earth-to-Moon launches.

3.4 Model 2: Reliability Model under Non-Ideal Operations

3.4.1 Motivation and Modeling Scope

The deterministic logistics model in Task 1 assumes perfect execution of all operations. In Task 2, we incorporate non-ideal conditions by modeling *irreversible failures* that permanently remove rockets from the active fleet. The objective is to quantify the distribution of completion time, failure counts, and the required replacement effort under each scenario, and to evaluate robustness relative to the Task 1 baseline.

3.4.2 Baseline Parameter Inheritance from Task 1

Task 2 is constructed as a stochastic extension of Task 1. Therefore, all deterministic quantities under ideal operation are inherited as baseline inputs for the non-ideal simulation, including: (i) nominal program durations $\{\tau_l\}_{l \in \mathcal{L}}$, (ii) nominal payload $q = Cap_{Rock}$, and (iii) the steady-state safety fleet threshold $I_{safe}^{(h)}$ derived in Section 3.1.5. In particular, the replacement controller in Task 2 uses the Task 1 fleet-sizing result as a reference level:

$$D(t) = (I_{safe}^{(h)} - I(t))_+.$$

All additional parameters introduced in Task 2 (e.g., failure probabilities p_{l6} , replacement latency δ , and robustness buffer $\Delta\tau$) represent deviations from the Task 1 baseline and are explored through sensitivity analysis.

3.4.3 Program Decomposition and Failure-State Definition

We decompose the rocket operations into the following five programs:

- (1) Rocket (empty): Earth launch site $\rightarrow$ Apex anchor (deployment),
- (2) Rocket (loaded): Earth surface $\rightarrow$ TLI staging orbit,
- (3) Rocket (loaded): (Apex or TLI) $\rightarrow$ Moon colony (deliver/land/unload),
- (4) Rocket (empty): Moon $\rightarrow$ Apex anchor (recovery/docking),
- (5) Rocket (empty): TLI $\rightarrow$ Earth launch site (return).

Let the operational set be $\mathcal{L} = \{1, 2, 3, 4, 5\}$.

To represent irreversible failures, we introduce an absorbing failure state 6. If a rocket experiences a critical malfunction during any program $l \in \mathcal{L}$, it enters state 6 and is permanently removed from the transportation system. The extended state space is

$$\bar{\mathcal{L}} = \mathcal{L} \cup \{6\}.$$

For each program $l \in \mathcal{L}$, let $p_{l6} \in [0, 1]$ denote the probability that an irrecoverable failure occurs during program l and triggers a transition into state 6.

3.4.4 Scenario-Dependent Nominal Program Rule

As in Task 1, each transportation scenario $h \in \{1, 2, 3\}$ induces a deterministic nominal program-transition rule $\phi_h : \mathcal{L} \rightarrow \mathcal{L}$:

Scenario 1 (Apex-based cycling after deployment):

$$\phi_1(1) = 3, \phi_1(3) = 4, \phi_1(4) = 3,$$

Scenario 2 (Earth-launched cycling):

$$\phi_2(2) = 3, \phi_2(3) = 5, \phi_2(5) = 2,$$

Scenario 3 (Hybrid: one loaded Earth launch then Apex cycling): $\phi_3(2) = 3, \phi_3(3) = 4, \phi_3(4) = 3.$

Programs not used in a given scenario are not entered under correct initialization. For numerical stability and to ensure row-stochastic transition matrices, we assign unused operational states to self-loops.

3.4.5 Absorbing Markov Transition Model for Rocket Failures

To incorporate non-ideal conditions, the next program is no longer deterministic. We model the program evolution of a rocket as a Markov process on $\bar{\mathcal{L}}$. Let

$$\bar{P}_{ll'}^h = \mathbb{P}(L^+ = l' \mid L = l, h), \quad l, l' \in \bar{\mathcal{L}},$$

denote the transition probability from state l to state l' under scenario h .

For each operational program $l \in \mathcal{L}$ such that $\phi_h(l)$ is defined, the rocket fails with probability p_{l6} and enters absorbing state 6; otherwise it proceeds to the nominal next program $\phi_h(l)$:

$$\bar{P}_{ll'}^h = \begin{cases} 1 - p_{l6}, & l' = \phi_h(l), \\ p_{l6}, & l' = 6, \\ 0, & \text{otherwise,} \end{cases} \quad \bar{P}_{66}^h = 1.$$

Explicit matrices (state order $(1, 2, 3, 4, 5, 6)$). Unused operational states are set to self-loops.

$$\bar{\mathbf{P}}^{(1)} = \begin{pmatrix} 0 & 0 & 1 - p_{16} & 0 & 0 & p_{16} \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 - p_{36} & 0 & p_{36} \\ 0 & 0 & 1 - p_{46} & 0 & 0 & p_{46} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\bar{\mathbf{P}}^{(2)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 - p_{26} & 0 & 0 & p_{26} \\ 0 & 0 & 0 & 0 & 1 - p_{36} & p_{36} \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 - p_{56} & 0 & 0 & 0 & p_{56} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\bar{\mathbf{P}}^{(3)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 - p_{26} & 0 & 0 & p_{26} \\ 0 & 0 & 0 & 1 - p_{36} & 0 & p_{36} \\ 0 & 0 & 1 - p_{46} & 0 & 0 & p_{46} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

3.4.6 Failure Probability Synthesis and Parameter Mapping

Event composition (per-operation). If a program consists of independent sub-events with failure probabilities $\{p_k\}$, then

$$P_{\text{succ}} = \prod_k (1 - p_k), \quad P_{\text{fail}} = 1 - P_{\text{succ}}.$$

This rule maps engineering failure modes into the Markov parameters p_{l6} .

Baseline mapping to the five programs. We map the dominant risk sources to each program as follows:

- Program 1: empty deployment to Apex (launch/insertion + rendezvous/docking).
- Program 2: loaded Earth launch to TLI (launch/injection).
- Program 3: loaded delivery leg to Moon (transfer + landing + unloading, aggregated).
- Program 4: empty recovery from Moon to Apex (ascent/return + rendezvous/docking).
- Program 5: empty return from TLI to Earth site (return/recovery).

Let p_{launch} denote per-launch failure probability for a mature heavy-lift-class launch vehicle, p_{dock} the probability of rendezvous/docking failure at the Apex, and p_{land} the lunar landing/ delivery failure probability [?, ?, ?].

Under the independence assumption between launch and docking sub-events, we set:

$$p_{16} = 1 - (1 - p_{\text{launch,empty}})(1 - p_{\text{dock}}), \quad p_{26} = p_{\text{launch,loaded}}, \quad p_{36} = p_{\text{land}},$$

$$p_{46} = 1 - (1 - p_{\text{return,moon} \rightarrow \text{apex}})(1 - p_{\text{dock}}), \quad p_{56} = p_{\text{return,earth}}.$$

In the baseline case, we approximate $p_{\text{launch,empty}} \approx p_{\text{launch,loaded}} \approx p_{\text{launch}}$, and keep $p_{\text{return,moon} \rightarrow \text{apex}}$ and $p_{\text{return,earth}}$ symbolic for sensitivity analysis.

Conservative alternative (correlated risk stress-test). If independence is deemed optimistic, a conservative bound can be used:

$$p_{16} = \max\{p_{\text{launch,empty}}, p_{\text{dock}}\}, \quad p_{46} = \max\{p_{\text{return,moon} \rightarrow \text{apex}}, p_{\text{dock}}\}.$$

Representative baseline magnitudes (for simulation). We adopt representative magnitudes as baseline inputs and perturb them over plausible ranges in sensitivity analysis:

$$p_{\text{launch}} = \mathcal{O}(10^{-2}), \quad p_{\text{land}} = \mathcal{O}(10^{-3}), \quad p_{\text{dock}} = \mathcal{O}(10^{-3} \sim 10^{-2}).$$

Table 5: Failure probability parameters for the absorbing Markov model.

Program l	Description	Absorbing failure prob. p_{l6}
1	Empty rocket: Earth $\rightarrow$ Apex (deploy + dock)	$1 - (1 - p_{\text{launch}})(1 - p_{\text{dock}})$
2	Loaded rocket: Earth $\rightarrow$ TLI (launch/injection)	p_{launch}
3	Loaded rocket: (Apex or TLI) $\rightarrow$ Moon (deliver/land)	p_{land}
4	Empty rocket: Moon $\rightarrow$ Apex (return + dock)	$1 - (1 - p_{\text{return,moon} \rightarrow \text{apex}})(1 - p_{\text{dock}})$
5	Empty rocket: TLI $\rightarrow$ Earth (return/recovery)	$p_{\text{return,earth}}$

3.4.7 Apex Inventory Coupling for Elevator-Supported Scenarios

For Scenario 1 and the steady phase of Scenario 3, payloads used in Program (3) must be supplied by the Space Elevator to the Apex anchor. We introduce an Apex inventory state $S(t)$ (ton), representing available cargo at the anchor for loading.

Effective elevator supply under non-ideal operations. We model non-ideal elevator operations via an effective supply rate Cap_{SE}^{eff} . A simple robustness model draws a downtime ratio $d \in [0, 1)$ per Monte Carlo trajectory and sets

$$Cap_{SE}^{\text{eff}} = (1 - d) Cap_{SE}.$$

(Alternatively, $Cap_{SE}^{\text{eff}}(t)$ can be modeled as a two-state up/down process; we use the trajectory-wise constant approximation for clarity.)

Inventory evolution between events. Let t be the global clock and t_{prev} the previous event time. Between events, inventory increases deterministically:

$$S(t) \leftarrow S(t_{\text{prev}}) + Cap_{SE}^{\text{eff}} \cdot (t - t_{\text{prev}}).$$

Loading constraint at the Apex. A loaded outbound delivery (Program 3) requires at least one payload unit q in inventory. When a rocket at the Apex is ready to start Program (3), it can only depart after sufficient inventory is available. If at time t we have $S(t) < q$, the earliest feasible loading time is

$$t_{\text{load}} = t + \frac{q - S(t)}{Cap_{SE}^{\text{eff}}}.$$

At time t_{load} , we deduct inventory and then execute Program (3):

$$S(t_{\text{load}}^+) \leftarrow S(t_{\text{load}}) - q.$$

This rule prevents phantom deliveries when the elevator supply is insufficient.

Direct-launch deliveries. For Program (2)→(3) deliveries sourced from Earth launches, we do not consume Apex inventory. Inventory coupling applies only to Apex-supported departures (Scenario 1 and the steady-state phase of Scenario 3).

3.4.8 Event-Driven Discrete-Event Simulation (DES)

We simulate the fleet evolution using a discrete-event mechanism. For each rocket r , let $L_r(t) \in \bar{\mathcal{L}}$ denote its current program/state at time t , and let $T_r(t) \geq t$ be its scheduled completion time.

Stochastic state transition at event times. When rocket r completes its current program $l = L_r(t) \in \mathcal{L}$ at time $T_r(t)$, its next state L_r^+ is sampled according to $\bar{\mathbf{P}}^{(h)}$:

$$\mathbb{P}(L_r^+ = l' \mid L_r = l) = \bar{P}_{ll'}^h, \quad l' \in \bar{\mathcal{L}}.$$

If $L_r^+ = 6$, the rocket is permanently removed from the system.

Time update rule with robustness buffer. Let τ_l be the nominal duration of program $l \in \mathcal{L}$. To reflect conservative scheduling, we inflate each duration by a buffer $\Delta\tau$:

$$\tau_l^{\text{robust}} = \tau_l + \Delta\tau, \quad l \in \mathcal{L}.$$

After the transition,

$$L_r(t^+) \leftarrow L_r^+, \quad T_r(t^+) \leftarrow \begin{cases} T_r(t) + \tau_{L_r^+}^{\text{robust}}, & L_r^+ \in \mathcal{L}, \\ +\infty, & L_r^+ = 6. \end{cases}$$

Setting $T_r = +\infty$ ensures failed rockets do not participate in future events.

Mass delivery accounting. Let $M(t)$ be the cumulative delivered mass to the Moon colony. Whenever a rocket completes Program (3), the delivered mass increases by payload q_r :

$$M(t^+) \leftarrow M(t) + q_r \quad \text{if Program 3 completes at time } t.$$

The simulation terminates once

$$M(t) \geq M_{\text{goal}}, \quad T^* = \inf\{t : M(t) \geq M_{\text{goal}}\}.$$

Global clock advancement. The global simulation time advances to the next event:

$$t \leftarrow \min_r T_r(t).$$

3.4.9 Closed-Loop Replacement with Latency and Launch Cadence

Irreversible failures reduce the active fleet size. Let

$$I(t) = \#\{r : L_r(t) \neq 6\}$$

denote the number of active rockets at time t . We issue replacement orders whenever $I(t) < I_{\text{safe}}^{(h)}$:

$$D(t) = (I_{\text{safe}}^{(h)} - I(t))_+.$$

Each replacement order becomes ready after a latency δ , but the actual insertion of new rockets is constrained by the maximum annual launch cadence f_{total} .

Launch-slot server (cadence constraint). Let t_{slot} be the next available launch-slot time (initialized $t_{\text{slot}} = 0$). When a replacement order becomes ready at time $t_{\text{ready}} = t + \delta$, its actual launch start time is

$$t_{\text{start}} = \max\{t_{\text{ready}}, t_{\text{slot}}\}, \quad t_{\text{slot}} \leftarrow t_{\text{start}} + \frac{1}{f_{\text{total}}}.$$

Scenario-dependent insertion state. A newly inserted rocket enters the scenario-specific first program:

$$l_0^{(1)} = 1, \quad l_0^{(2)} = 2, \quad l_0^{(3)} = 2.$$

At t_{start} , we create a new rocket r_{new} with

$$L_{r_{\text{new}}}(t_{\text{start}}^+) = l_0^{(h)}, \quad T_{r_{\text{new}}}(t_{\text{start}}^+) = t_{\text{start}} + \tau_{l_0^{(h)}}^{\text{robust}}.$$

Replacement accounting. Let N_{launch} be the total number of replacement launches executed in Task 2. Each insertion increments:

$$N_{\text{launch}} \leftarrow N_{\text{launch}} + 1.$$

If Task 2 requires cost reporting, the additional launch cost term can be included as $C_{\text{launch}} \cdot N_{\text{launch}}$.

3.4.10 DES Routine with Inventory and Replacement

We summarize the complete DES logic:

Initialization. Set $t = 0$, $t_{\text{prev}} = 0$, $M(0) = 0$, choose scenario h , initialize $I(0)$ rockets and their initial programs, set $S(0) = 0$, choose Cap_{SE}^{eff} (Scenario 1/3), and set $t_{\text{slot}} = 0$.

Main loop. Repeat until $M(t) \geq M_{\text{goal}}$:

1. Advance time to next event $t \leftarrow \min_r T_r(t)$ and update inventory $S(t) \leftarrow S(t_{\text{prev}}) + Cap_{SE}^{\text{eff}}(t - t_{\text{prev}})$, $t_{\text{prev}} \leftarrow t$.

2. Let r be the rocket completing Program $l = L_r(t)$. Sample L_r^+ from $\bar{\mathbf{P}}^{(h)}$.
3. If $L_r^+ = 6$, set $T_r(t^+) = +\infty$ and issue $D(t)$ replacement orders; each order is inserted via the launch-slot server after latency δ .
4. If $L_r^+ \in \mathcal{L}$, schedule the next completion time:
 - If $L_r^+ = 3$ and the departure is Apex-supported (Scenario 1 or steady Scenario 3), enforce loading: if $S(t) < q$, delay to $t_{\text{load}} = t + (q - S(t))/Cap_{SE}^{\text{eff}}$; deduct q and set $T_r = t_{\text{load}} + \tau_3^{\text{robust}}$.
 - Otherwise set $T_r = t + \tau_{L_r^+}^{\text{robust}}$.
5. Delivery accounting: whenever Program 3 completes, update $M(t^+) = M(t) + q$.

This DES guarantees: (i) failed rockets never re-enter; (ii) $M(t)$ is monotone non-decreasing; (iii) the global time t strictly increases; (iv) Apex-supported deliveries cannot exceed cumulative elevator supply.

3.4.11 Monte Carlo Protocol, Outputs, and Verification Checks

Because T^* and failure counts are random under the absorbing Markov dynamics, we estimate performance statistics via Monte Carlo simulation. We run N_{MC} independent trajectories, each simulated until $M(t) \geq M_{\text{goal}}$.

Key outputs. For each trajectory n , record:

$$T^{*(n)}, \quad N_{\text{fail}}^{(n)}, \quad N_{\text{launch}}^{(n)}, \quad \max_t (I_{\text{safe}}^{(h)} - I(t))_+,$$

and compute sample mean and standard deviation:

$$\bar{T}^* = \frac{1}{N_{\text{MC}}} \sum_{n=1}^{N_{\text{MC}}} T^{*(n)}, \quad s_T = \sqrt{\frac{1}{N_{\text{MC}} - 1} \sum_{n=1}^{N_{\text{MC}}} (T^{*(n)} - \bar{T}^*)^2}.$$

A 95% confidence interval for $\mathbb{E}[T^*]$ is approximated by

$$\bar{T}^* \pm 1.96 \frac{s_T}{\sqrt{N_{\text{MC}}}}.$$

Implementation verification checks. To ensure correctness of the DES implementation, we enforce:

1. If a rocket enters state 6, then $T_r(t^+) = +\infty$ and it never contributes to delivery again;
2. The cumulative delivered mass $M(t)$ is monotone non-decreasing;
3. The global time t is strictly increasing across events;
4. For Apex-supported scenarios, the total payload departures from Apex do not exceed the cumulative inventory inflow.

Remark on parameter sourcing. All baseline probabilities are treated as scenario inputs; their baseline magnitudes are motivated by reliability and docking literature ^[?, ?, ?] and then perturbed to quantify sensitivity and robustness.

4 Validating the Model

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5 Conclusions

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6 A Summary

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7 Evaluate of the Mode

8 Strengths and weaknesses

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placemat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

8.1 Strengths

- **Applies widely**

This system can be used for many types of airplanes^[1,2], and it also solves the interference during the procedure of the boarding airplane, as described above we can get to the optimization boarding time. We also know that all the service is automate.

- **Improve the quality of the airport service**

Balancing the cost of the cost and the benefit, it will bring in more convenient for airport and passengers. It also saves many human resources for the airline.

-

References

[1] D. E. KNUTH The T_EXbook the American Mathematical Society and Addison-Wesley Publishing Company, 1984-1986.

[2] Lamport, Leslie, L^AT_EX: "A Document Preparation System", Addison-Wesley Publishing Company, 1986.

[3] <http://www.latexstudio.net/>

[4] <http://www.chinatex.org/>

Appendices

Appendix A First appendix

Aliquam lectus. Vivamus leo. Quisque ornare tellus ullamcorper nulla. Mauris portitor pharetra tortor. Sed fringilla justo sed mauris. Mauris tellus. Sed non leo. Nullam elementum, magna in cursus sodales, augue est scelerisque sapien, venenatis congue nulla arcu et pede. Ut suscipit enim vel sapien. Donec congue. Maecenas urna mi, suscipit in, placerat ut, vestibulum ut, massa. Fusce ultrices nulla et nisl.

Here are simulation programmes we used in our model as follow.

Input matlab source:

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
```

```
    if pab(i)<0.4
        aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

Appendix B Second appendix

some more text **Input C++ source:**

```
//=====
// Name      : Sudoku.cpp
// Version    : a.0
// Copyright  : Your copyright notice
// Description: Sudoku in C++.
//=====

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
        for(int y = 0; y < 9; y++){
            cout << table[x][y] << " ";
        }

        cout << endl;
    }
    cout << "Python is a best language";

    return 0;
}
```

Report on Use of AI

OpenAI-GPT5-Thinking

Query1 : prompt1

Output1:

AI Output